The first graph depicts a hypothetical movement of Bombay Stock. Exchange Sensitive index which is an indicator of the average price movement of a representative set of stocks. The second graph depicts the averaged price of all automobile manufacturing companies. Which one of the following is a valid inference ?

(a) The market in unstable because prices are continuously fluctuating
(b) Automobile shares are more stable than shares in general
(c) There was a major political change in June/July
(d) Automobile shares have been steadily improving in price rather than being affected by major market movements

Ans. (a) : From the graph we know that the first graph depicts a hypothetical movement of Bombay Stock Exchange sensitive index which is an indicator of the average price movement of a representative set of stocks.
And the second graph depicts the averaged price of all automobile manufacturing companies, So Automobile shares have been steadily improving in price rather than being affected by major market movements.

80. The table below compares apparel exports of various countries between the years 1980 and 1990.

Apparel Exports of Various Countries (in billions)

Country		
Hong Kong	4.98	15.41
Italy	4.58	11.84
China	1.63	9.67
S.Korea	2.95	7.88
Thailand	0.27	3.33
India	0.59	2.50
Indonesia	0.10	1.65
Malaysia	0.15	1.32
Total	15.25	53.60

Which one of the following statements is not true ? (Restricting consideration to countries listed in the table only)

(a) Thailand has overtaken India in apparel export performance between 1980 and 1990
(b) Total apparel export from the countries listed has increased (in dollar terms) between 1980 and 1990 but part of this is likely to be due to global inflation
(c) India's share of the apparel export market has increased
(d) Maximum improvement in apparel export performance has been achieved by Indonesia

Ans. (b) : The export for all given eight countries has increased between 1980 and 1990, but the reason behind this increment is not mentioned anywhere in question. Hence, it would be wrong to say that this is due to global inflation. Therefore, statement given in option (b) is not true.

PHYSICS

81. The colour of a star is an indication of its

(a) distance from the sun
(b) luminosity
(c) distance from the earth
(d) temperature

Ans. (d) : The colour of a star is an indication of its temperature, and it can also suggest the star's age. Astronomers have long used stars colour to identify their temperature, and they created a classification system called spectral classes to do so, which places stars into a category based on their temperature.

82. The basic reason for the extraordinary sparkle of a suitably cut diamond is that

(a) it has a very high transparency
(b) it has a very high refractive index
(c) it is very hard
(d) it has welldefined cleavage planes

Ans. (b) : The refractive index of diamond ($\mu = 2.42$) is very high. Due to which its angle of incidence is greater than the critical angle. This results in total internal reflection of the light entering it, which is the fundamental factor in the brilliance of the diamond.

83. A corked bottle full of water when frozen will break because

(a) the bottle contracts on freezing
(b) the volume of water decreases on freezing
(c) the volume of water increases on freezing
(d) glass is a bad conductor of heat

Ans. (c) : A corked bottle full of water when frozen will break because the volume of water increases on freezing.

84. Which one of the following combinations of aperture and shutter speed of a camera will allow the maximum exposure ?

(a) f22, 1/60 (b) f16, 1/125
(c) f8, 1/250 (d) f5.6, 1/1000

Ans. (a) : The exposure of the camera depends on the area and time of the aperture of the camera, so the exposure will be directly proportional to $d^2 \times t$.

85. The speed of light will be minimum while passing through

(a) glass (b) vacuum
(c) water (d) air

Ans. (a) : Out of glass, vacuum, water and air, the velocity of light in glass will be the least. The velocity of light in vacuum is 3 lakh km/sec & in glass 2 lakh km/se.

86. The colour of an opaque object is due to the colour it

(a) absorbs (b) refracts
(c) reflects (d) scatters

Ans. (c) : The colour of an opaque object is due to the light waves that it reflects. The rest of the wavelengths get reflected from the surface of the object. These reflected wavelengths of light reach our eyes and we perceive them.

87. The pitch of the voice of women is in general

(a) marginally lower than that of men
(b) higher than that of men
(c) much lower than that of men
(d) the same as that of men

Ans. (b) : Generally, the pitch of women's voice is higher than that of men, due to which the voice of women is thin and sweet.

88. In an electronic watch, the component corresponding to pendulum of a pendulum clock is a

(a) transistor (b) crystal oscillator
(c) diode (d) balance wheel

Ans. (b) : The crystalline oscillator is the same component (instrument) in both the electronic clock and the pendulum clock, due to this device, the electronic and pendulum clocks indicates the correct time.

CHEMISTRY

89. One mole of hydrogen gas burns in excess of oxygen to give 290 kJ of heat. What is the amount of heat produced when 4g of hydrogen gas is burnt under the same conditions ?

(a) 145 kJ (b) 290 kJ
(c) 580 kJ (d) 1160 kJ

Ans. (c) : One mole of hydrogen gas = 2 gm
$\because$ when 4 gm of hydrogen gas burnt the
energy = $290 \times 2 = 580$ kj

90. The halflife of a radio active element is 5 years. The fraction of the radioactive substance that remains after 20 years is

(a) 1/2 (b) 1/4
(c) 1/8 (d) 1/16

Ans. (d) : Let A_i be initial concentration
$t_{1/2} = 5$ years

$\therefore$ After 5 years, concentration remaining is $\frac{A_i}{2}$

For first order reaction, we know

$$k = \text{constant} = \frac{0.693}{t_{1/2}}$$

$$\Rightarrow k = \frac{0.693}{5}$$

Now, let A_{20} be the concentration after 20 years.

Then, $k = \frac{2.303}{t}\log_{10}\left(\frac{A_i}{A_t}\right)$

$$\Rightarrow \frac{0.693}{5} = \frac{2.303}{20}\log_{10}\left(\frac{A_i}{A_{20}}\right)$$

$$\Rightarrow \log_{10}\left(\frac{A_i}{A_{20}}\right) = \frac{0.693\times 20}{5\times 2.303}$$

$$\Rightarrow \left(\frac{A_i}{A_{20}}\right) = 16 \text{ (After solving)}$$

$\Rightarrow A_{20} = \frac{A_i}{A_{16}}$ so, fraction remaining $= \frac{1}{16}$

91. Monazite is an ore of

(a) zirconium (b) thorium
(c) titanium (d) iron

Ans. (b) : Monazite is an ore of thorium. In a typical acid digestion process, monazite is cracked using sulfuric acid at 230^0C for 4 h. The process is followed by a selective precipitation process using ammonium hydroxide (NH_4OH), sodium hydroxide (NaOH), and hydrogen peroxide (H_2O_2) to separate thorium from rare earth elements and uranium.

92. Galvanised iron sheets are protected from rusting due to the presence of a layer of

(a) lead (b) chromium
(c) zinc (d) tin

Ans. (c) : Galvanised iron sheets are protected from rusting due to the presence of a layer of zinc.
Galvanizing is a process of coating iron or steel with zinc in order to provide greater protection against corrosion for the iron.
The zinc coating on galvanized iron and steel develops a natural carbonate on its surface by exposure to the atmosphere and by the action of rain water.